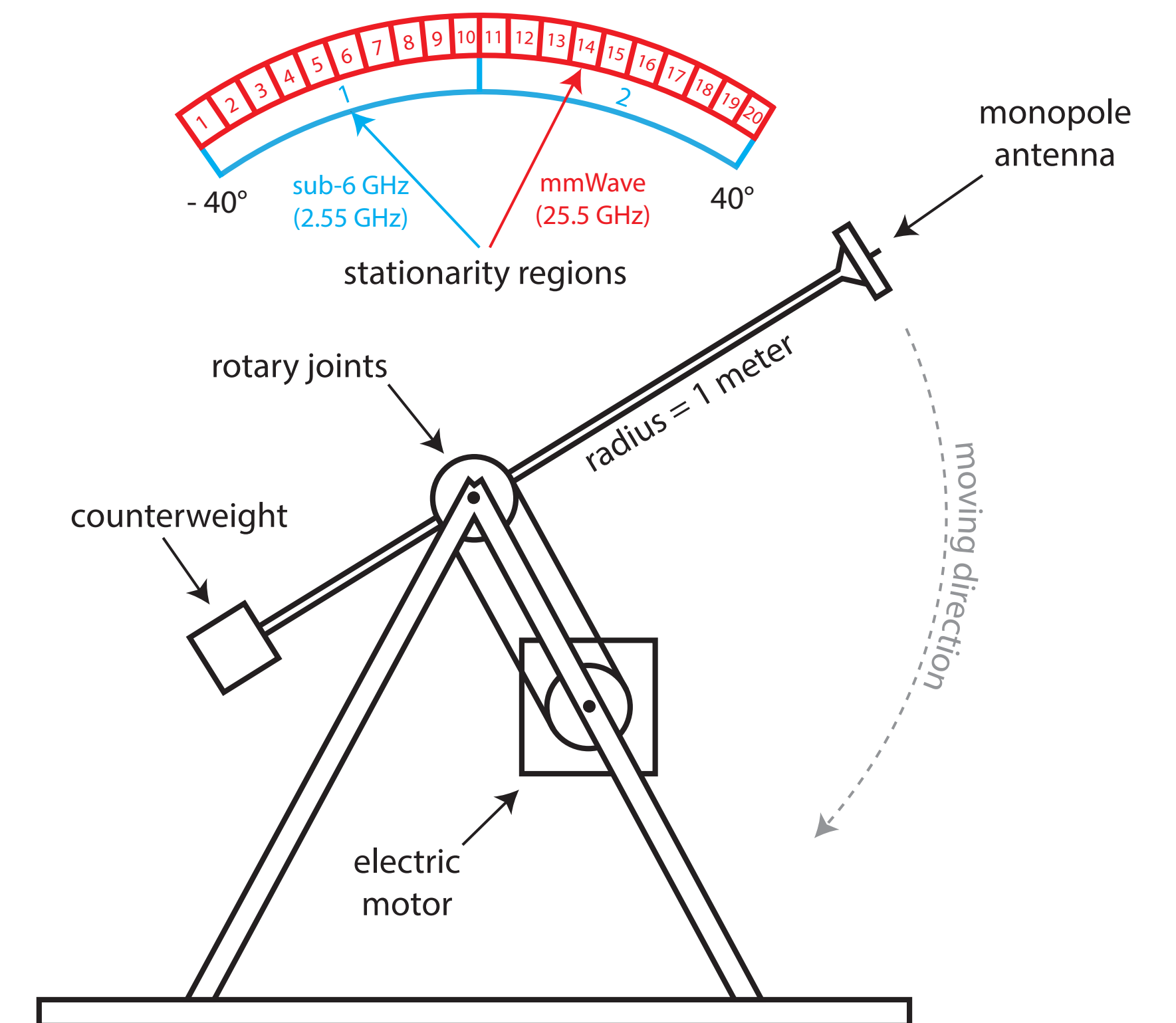


① Motivation

- High data transmission rates required
- Understand differences in terms of propagation channel characteristics between different bands
- Compare sub-6 GHz and millimeter wave (mmWave) frequency bands in a fair manner
- Problem:** conducting comparative multi-band measurements in high-mobility scenarios
 - Same time → antenna position offset
 - Same position → repeatability lost
 - Altered channel statistics

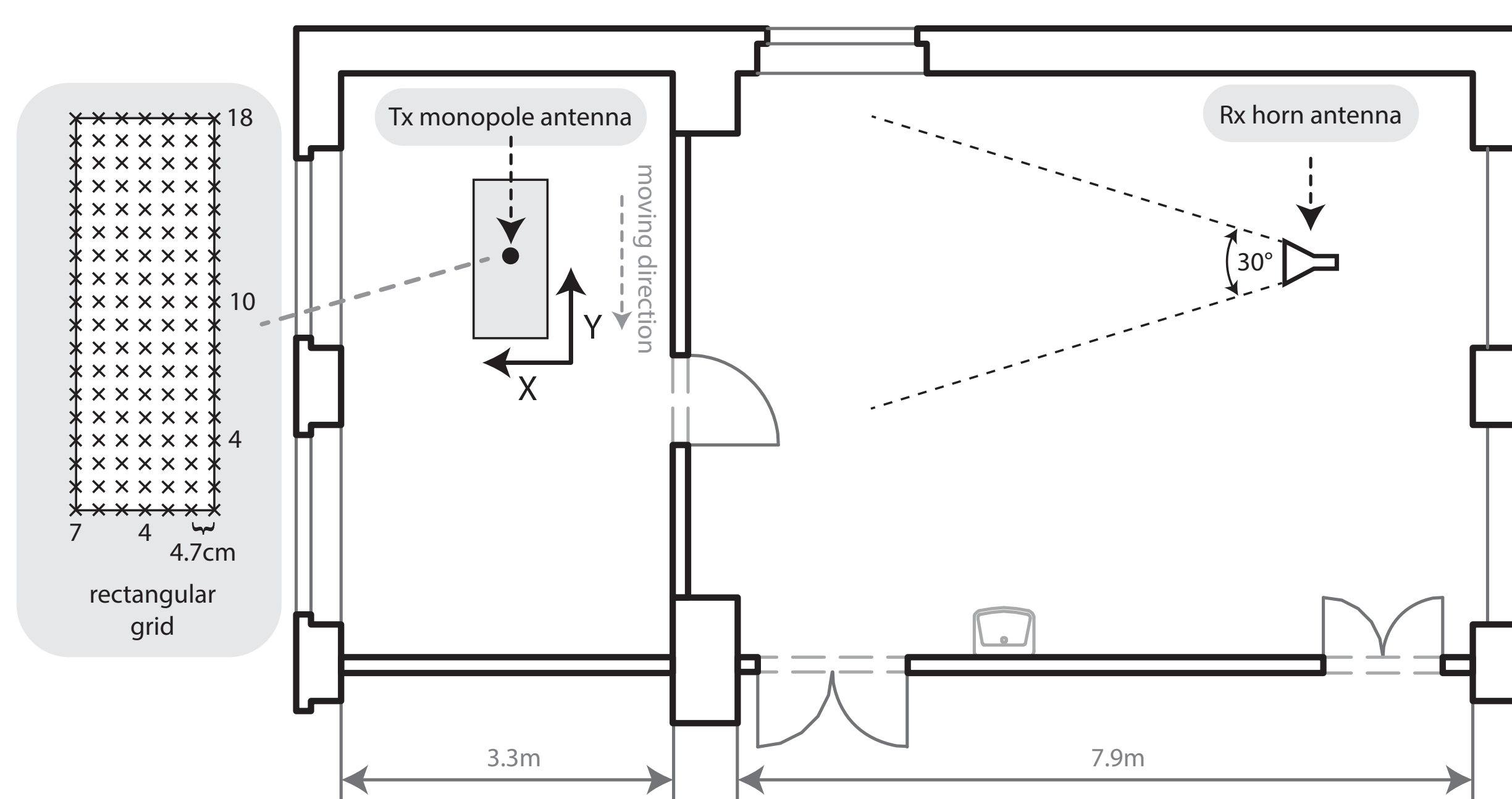
② Measurement Setup

- Operates in sub-6 GHz and mmWave bands
- Rotary unit spins a transmit antenna with a constant speed of up to 400 km/h
- Allows for repeatable multi-band high-speed wireless channel measurements
- Channel sounding using orthogonal frequency division multiplexing (OFDM)
- Identical antenna positions (paths) for different frequency bands



③ Measurement Campaign

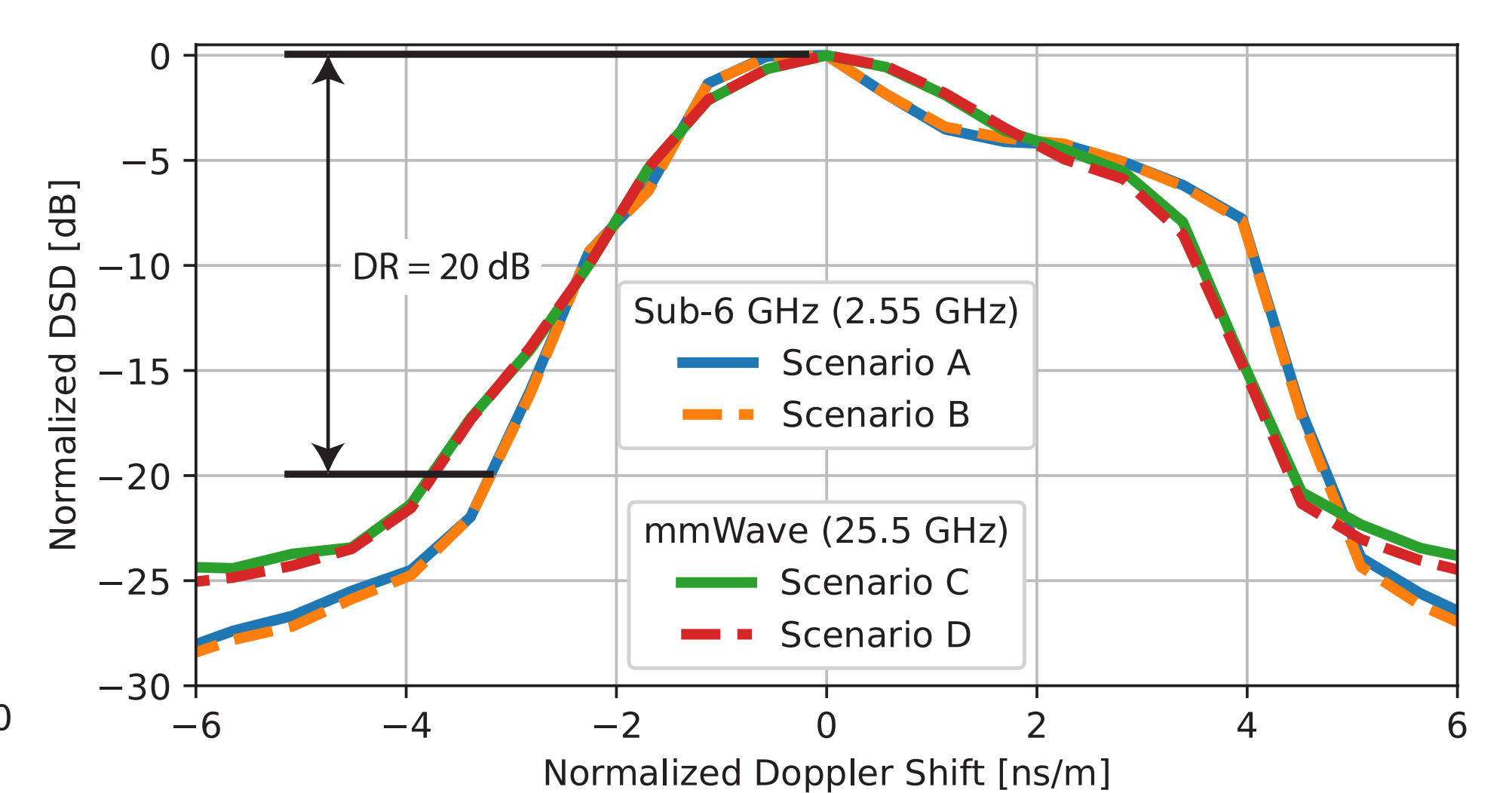
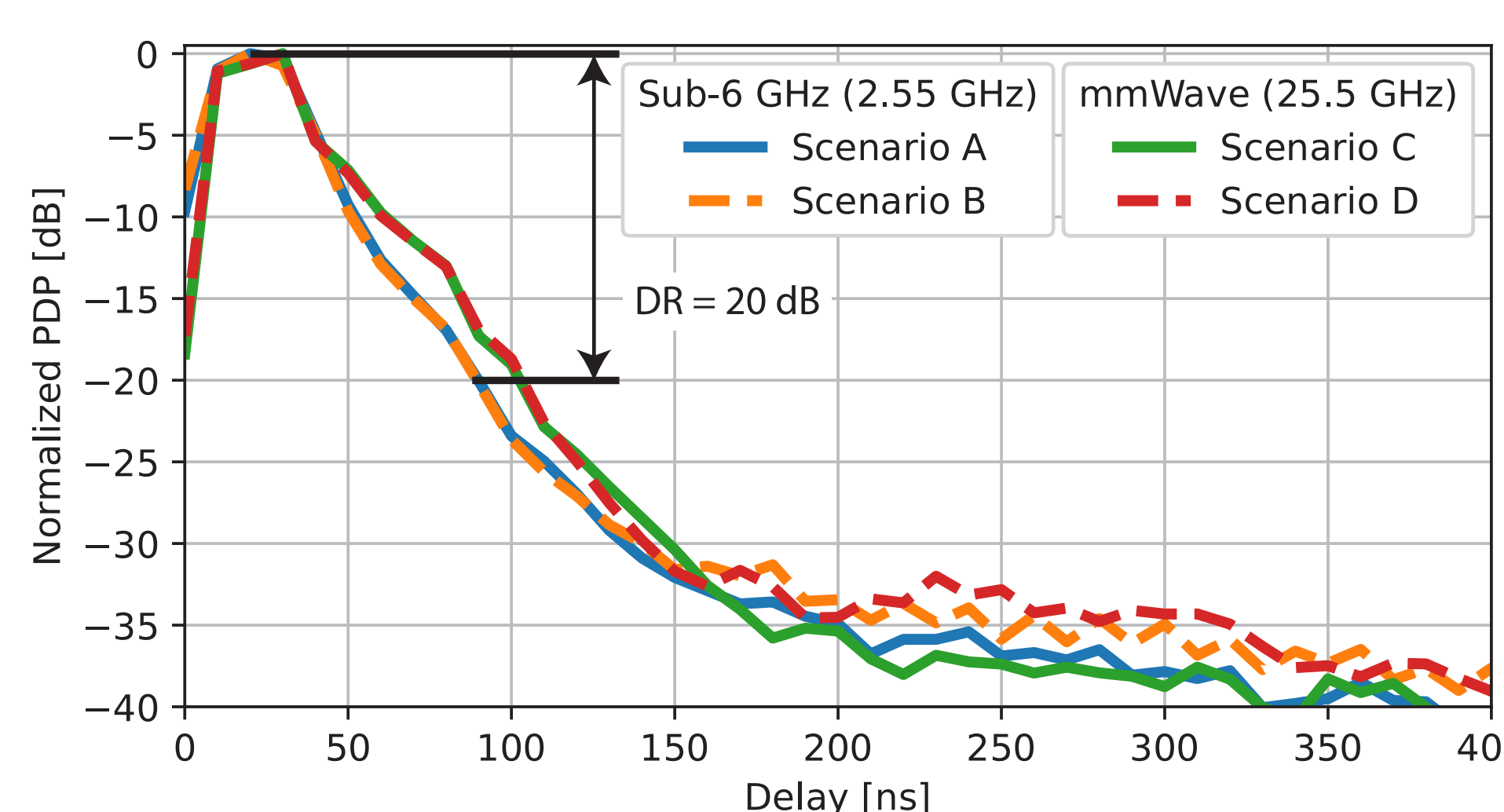
- Wireless channel measurements conducted in controlled indoor laboratory environment
 - Moving Tx and static Rx antenna
 - Static fading environment
- Measurements conducted at
 - Center frequencies of 2.55 and 25.5 GHz
 - Velocities of 50 and 100 km/h
 - 126 Different spatial transmitter positions
- Time-variant channel transfer function (CTF)



Parameter	Value			
Bandwidth B [MHz]	100			
Delay Resolution $\Delta\tau$ [ns]	10			
Number of Snapshots N	500			
Symbols per Snapshot N_{sym}	100			
Number of x-axis Positions N_x	7			
Number of y-axis Positions N_y	18			
Measured Arc Segment	$-40^\circ \dots 40^\circ$			
	A	B	C	D
Carrier Frequency f_c [GHz]	2.55	2.55	25.5	25.5
Wavelength λ [cm]	11.7	11.7	1.17	1.17
Transmitter Velocity v_{Tx} [km/h]	50	100	50	100
Number of Subcarriers K	200	100	200	100
Subcarrier Spacing Δf [kHz]	500	1000	500	1000
Symbol Duration t_s [μ s]	2	1	2	1
Snapshot Duration t_{snap} [μ s]	200	100	200	100
Measurement Duration t_m [ms]	100	50	100	50
Max. Doppler Shift ν_{max} [Hz]	118.1	236.2	1181	2362

④ Measurement Evaluation

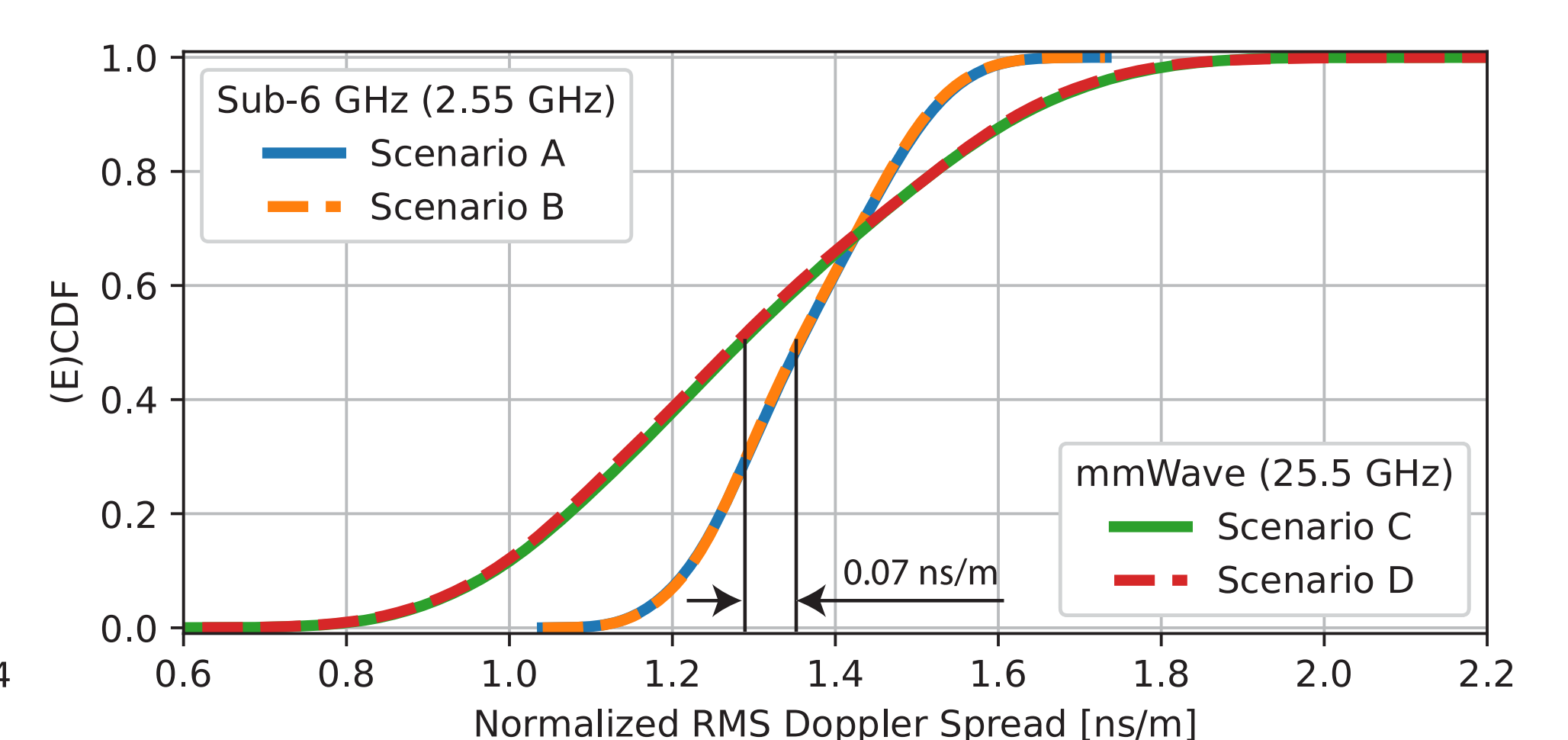
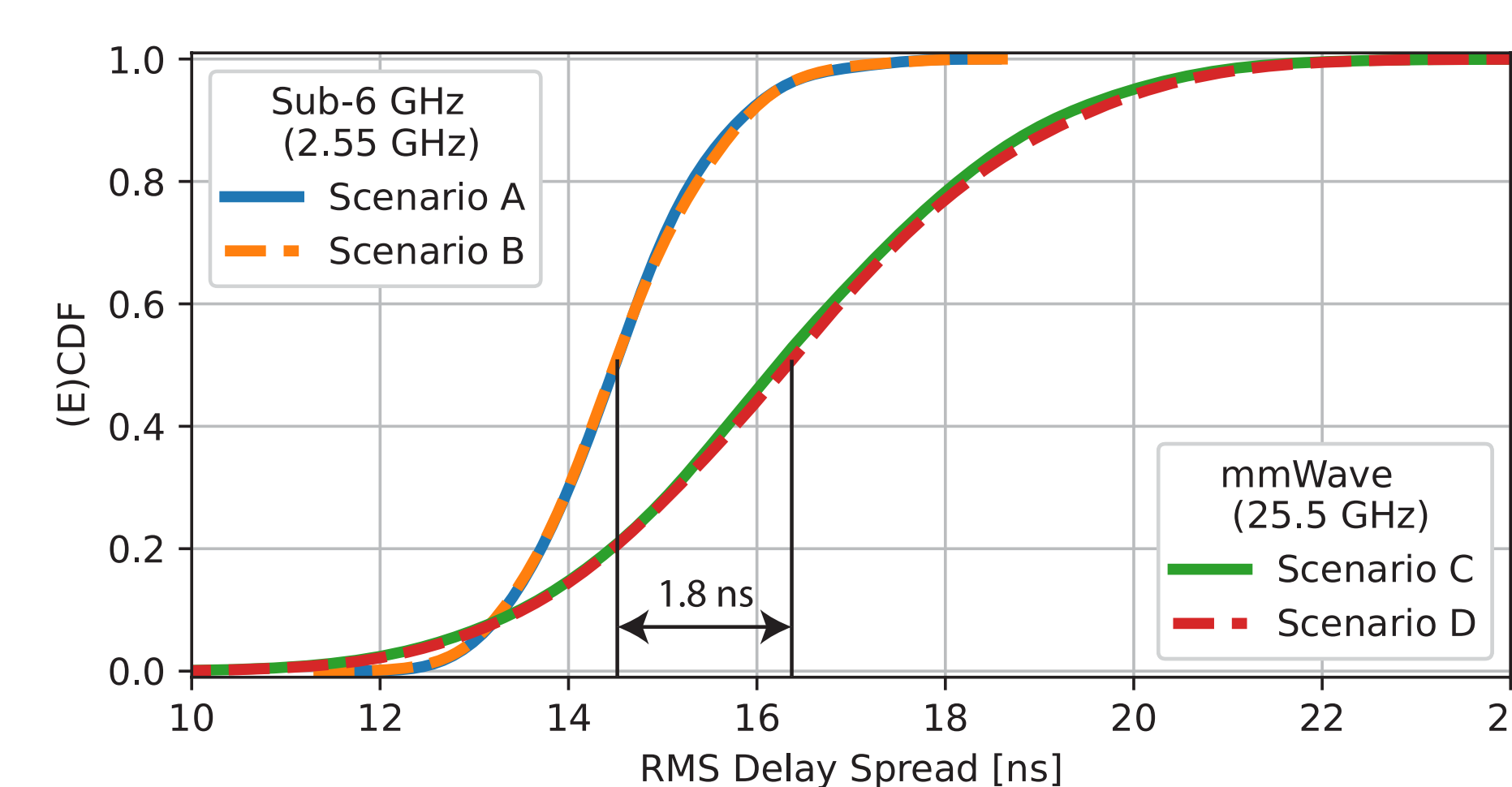
- Estimate local scattering function (LSF)
- Stationarity assumed over 6λ of motion
- Calculate power delay profile (PDP) and Doppler spectral density (DSD)
- Adjustments to allow better comparison
- RMS delay spread and normalized Doppler spread



⑤ Results and Discussion

- Cumulative distribution function (CDF)
- Invariance with regard to the transmitter's velocity
- Higher standard deviation for mmWave channels
 - Stationarity region of 6λ spatially smaller
 - More mmWave stationarity regions
 - Distance of 4λ between spatial positions
- Mean values similar for different frequency bands
 - RMS delay spread: 12% difference
 - norm. RMS Doppler spread: 5% difference
 - RMS Doppler shift 40% less than maximum

⑥ RMS Delay Spread and Normalized RMS Doppler Spread



⑦ Summary

- Proposed testbed can reproduce an experiment with minor uncertainties
- Small differences on channel spreads → almost the same propagation conditions at different frequency bands